

SAR2Earth: A SAR-to-EO Translation Dataset for Remote Sensing Applications

Supplementary Material

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Abstract

This document provides supplementary information that is not detailed in the main paper due to space constraints, including Dataset Details and Model Pipeline (Section A), Details of Land Cover Segmentation (Section B), Additional Qualitative Results (Section C), and Limitations and Border Impact (Section D).

A. Dataset Details and Model Pipeline

A.1. Data splits

The SAR2Earth dataset is divided into train, validation, and test sets based on spatial regions to ensure unbiased evaluation and consistency. The splits are defined as 70% for training, 10% for validation, and 20% for testing, considering the spatial distribution of each region to ensure robust generalization. This structured splitting approach ensures robust performance evaluation in SAR-to-EO translation tasks.

A.2. Co-Registration

To address the challenge of SAR-EO registration, various approaches can be considered, including deep learning-based methods and coordinate system alignment. In our study, we applied state-of-the-art data-driven matchers, such as MatchAnything [1] and GIM [6], which are known for their strong generalization across diverse domains. MatchAnything is a large-scale pre-trained model designed for cross-modality matching tasks, while GIM is a video-based self-supervised matcher that leverages diverse visual data. These models were selected based on the expectation that their generalization capabilities could address the difficulties of aligning heterogeneous modalities like SAR and EO images.

However, despite their promising performance on other cross-domain tasks, both methods struggled with the substantial modality gap between SAR and EO, such as the speckle noise and texture-based features of SAR versus the

color and intensity-based characteristics of EO. As shown in Figure 1, MatchAnything shows reasonable matching performance in the first row, but it fails to provide consistent results across different patches. Similarly, GIM produces unreliable correspondences and fails to generate valid warped SAR outputs, revealing limitations in applying current data-driven matching techniques to our dataset.

Given these challenges, we adopted a coordinate system registration approach, reprojecting both datasets into a unified system (e.g., WGS84). This strategy minimizes alignment errors and provides a more robust and generalizable framework for ensuring spatial consistency across the modalities.

A.3. Label maps and metadata

The label maps and metadata in the SAR2Earth dataset offer vital information for interpreting and analyzing the data. The label maps represent topological information, categorizing land cover classes derived from OpenStreetMap (OSM) data. Each label map is spatially aligned with its corresponding SAR and EO image pair, facilitating comprehensive spatial and semantic analysis.

The metadata includes fields such as geographic coordinates (based on the image center), ground sampling distance (GSD), and acquisition dates for both SAR and EO images. These components provide detailed context for each image pair, supporting diverse remote sensing applications. Figure 2 presents examples of dataset components, including label maps and metadata, for better understanding.

A.4. Comparison to Existing Datasets

Table 1 provides a comparative overview of the proposed SAR2Earth dataset and existing multi-modality datasets that include SAR and EO imagery. It highlights differences in modality, resolution, data types (e.g., Pan-sharpened, Multi-Spectral, RGB), application tasks (Data Fusion, Building Extraction, Image-to-Image Translation), the geographical distribution of regions, as well as temporal coverage. This comparison illustrates that SAR2Earth

Modality	Dataset	Year	Resolution (m)	Types	Task	Domain Urban / Rural	Regions	Temporal
SAR	SEN12 [5]	2018	10	VV	DF	Both	≥ 80	Dec 2016 – Nov 2017
	SARptical [8]	2018	1	N/A	DF	Urban	1	Jan 2009 – Dec 2013
	SpaceNet6 [7]	2020	0.5	HH, HV, VH, VV	BE	Urban	1	Aug 2019
	MAVIC-T [3]	2023	N/A	N/A	I2I	N/A	1	Aug 2008
	MAGIC [4]	2024	N/A	N/A	I2I	N/A	4	N/A
	SAR2Earth (ours)	2024	0.3 – 0.6	HH	I2I	Both	8	Feb 2021 – Sep 2024
EO	SEN12 [5]	2018	10	RGB	DF	Both	≥ 80	Dec 2016 – Nov 2017
	SARptical [8]	2018	0.2	RGB	DF	Urban	1	Jan 2009 – Dec 2013
	SpaceNet6 [7]	2020	0.5	P, MS, PS	BE	Urban	1	Aug 2019
	MAVIC-T [3]	2023	N/A	RGB	I2I	N/A	1	Aug 2008
	MAGIC [4]	2024	N/A	RGB	I2I	N/A	4	N/A
	SAR2Earth (ours)	2024	0.15 – 0.6	RGB	I2I	Both	8	Nov 2016 – Apr 2024

Table 1. Comparison of SAR2Earth dataset with existing SAR and EO multi-modality datasets. P, MS, and PS represent Panchromatic (P), Multi-Spectral (MS), and Pan-Sharpned (PS) image types. Tasks (DF, BE, I2I) denote Data Fusion, Building Extraction, and Image-to-Image translation.

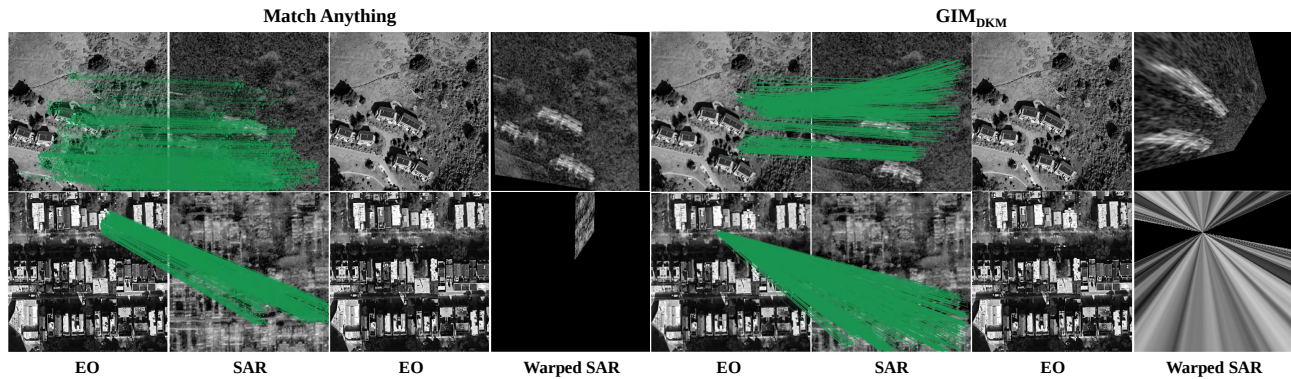


Figure 1. Results of state-of-the-art data-driven co-registration methods.

is characterized by its combination of high-resolution data (0.15–0.6m) and broad geographic coverage (8 areas), supporting both urban and rural domains and covering an extended temporal range (November 2016 – April 2024), which collectively offer advantages in diversity and resolution compared to existing datasets.

A.5. FLOW Chart

Figure 3 shows the pipeline of our SAR-to-EO approach along with image samples from each step. As depicted in the figure, all our models are connected sequentially, and during actual inference, only SAR images are used.

B. Details of Land Cover Segmentation

We utilized the official code¹ and dataset provided by OpenEarthMap (OEM). The model was trained on the entire dataset, including the xView2 Dataset, while adhering to the data splits defined in the official code. To adapt the training

¹https://github.com/bao18/open_earth_map

process for our target dataset, the images were resized to 256×256 and converted to grayscale before training.

B.1. Class Mapping

Due to the differences between the labels in our dataset and those provided by the OEM dataset used for training, we performed a class mapping process. The detailed mapping scheme is as follows:

- **Cropland:** Includes *allotments*, *farmland*, *greenhouse horticulture*, *orchard*, *plant nursery*, and *vineyard*.
- **Pavement:** Includes *apron*, and *runway*.
- **Bareland:** Includes *bare*, *bare rock*, and *sand*.
- **Water:** Includes *bay*, *salt pond*, and *water*.
- **Buildings:** Includes *construction*, *commercial*, *industrial*, *residential*, *retail* and *farmyard*.
- **Grass:** Includes *flowerbed*, *grass*, *grassland*, *heath*, *meadow*, *national park*, and *scrub*.
- **Tree:** Includes *forest* and *wood*.

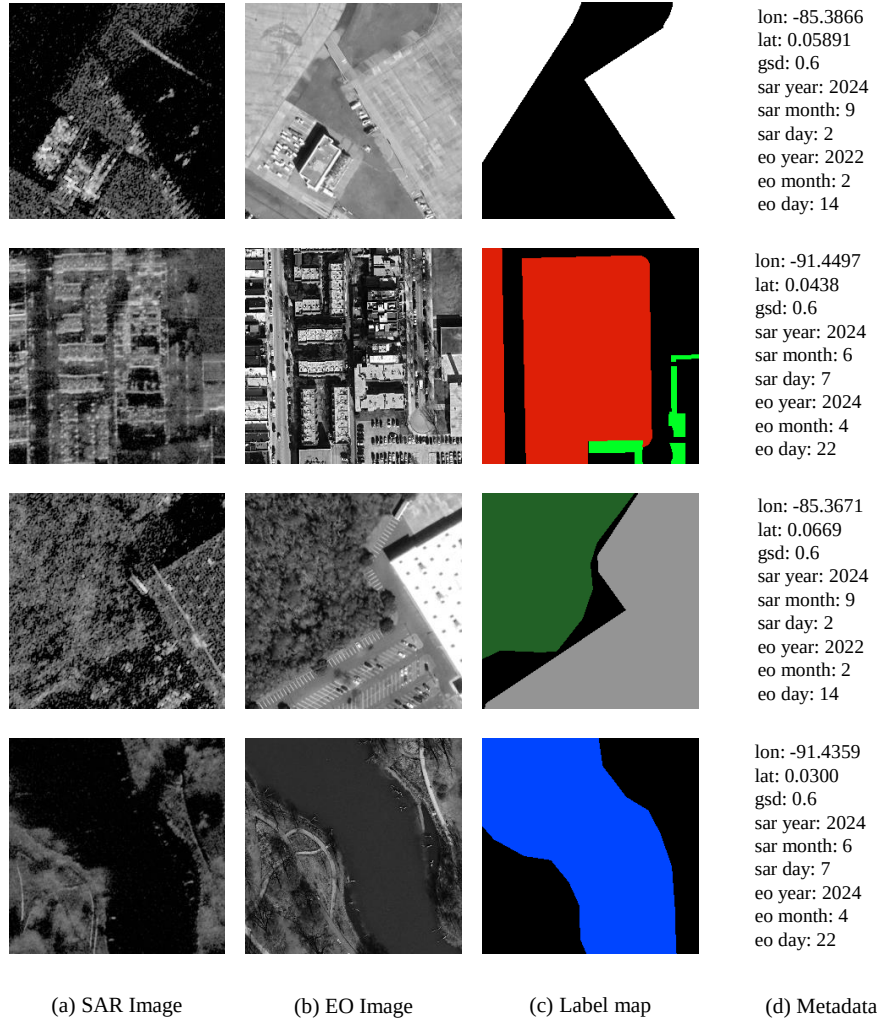


Figure 2. Overview of the SAR2Earth dataset components.

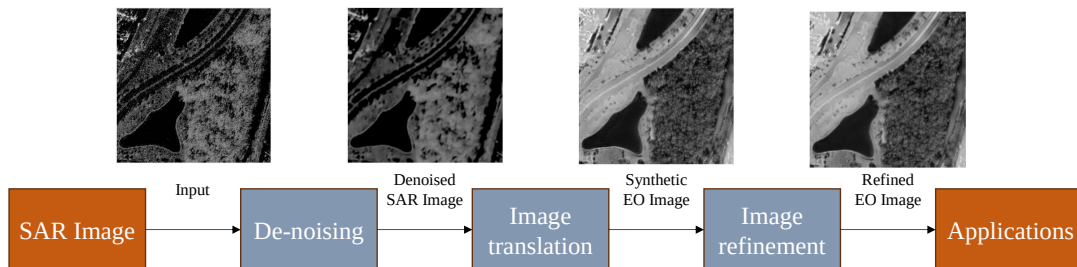


Figure 3. Overview of the SAR-to-EO translation pipeline.

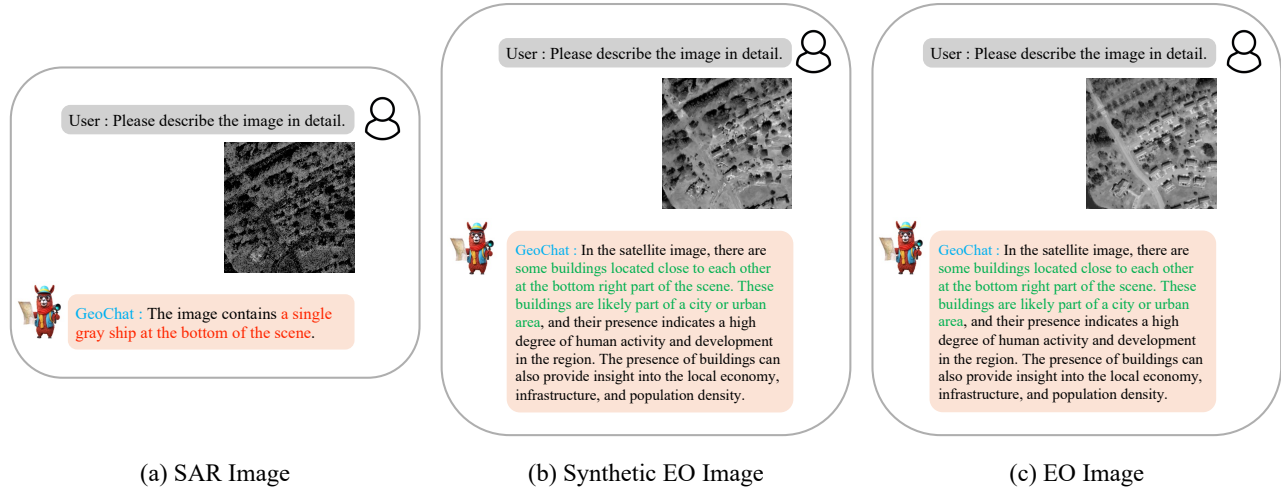


Figure 4. Interpretation differences between SAR and EO images. The SAR image (a) leads to an incorrect identification of objects in the scene, such as mislabeling a ship, while the synthetic EO (b) and real EO (c) images accurately capture key urban features, including clusters of buildings.

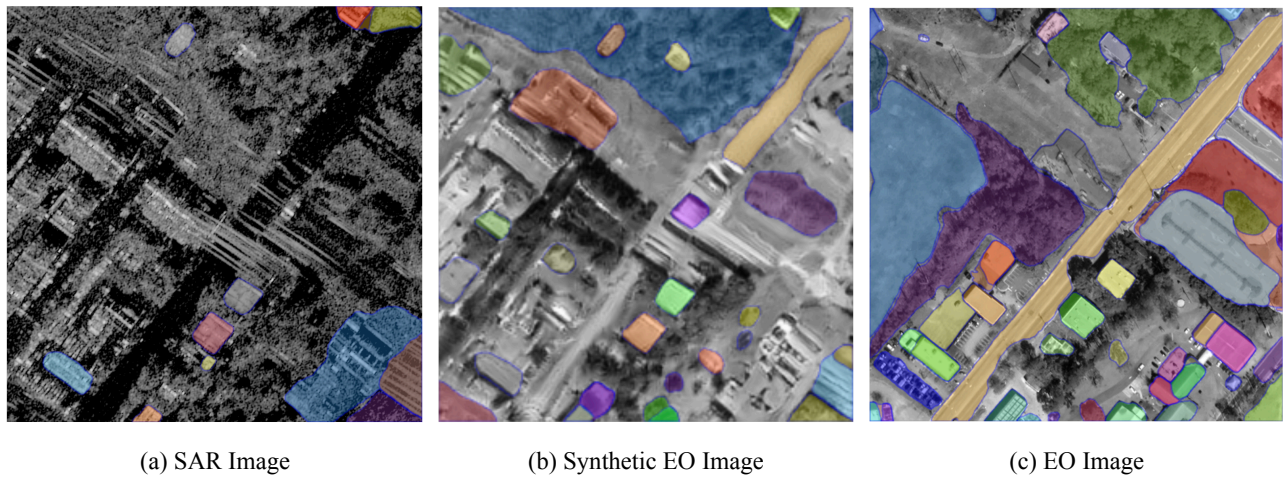


Figure 5. Segmentation results using SAM (Segment Anything Model) for different modalities.

C. Additional Qualitative Results

This section presents additional qualitative results that can not be included in the main manuscript.

C.1. Image interpretation

Figure 4 presents the results of image interpretation. As shown in the figure, GeoChat, which is trained exclusively on large-scale EO datasets, naturally fails to provide meaningful outputs for SAR images due to the absence of SAR data in its training. This result highlights that directly applying EO-trained models to SAR inputs is fundamentally limited.

However, by leveraging our dataset to perform SAR-to-

EO image translation, it becomes possible to utilize existing EO-trained models like GeoChat without the need to collect and train on a massive SAR dataset. This demonstrates the practical advantage of SAR-to-EO translation in bridging the gap between SAR and EO domains for downstream tasks.

C.2. Segment Anything

We tested the Segment Anything model [2] on SAR, SynEO, and EO images. As shown in Figure 5, the model struggled to perform effectively on SAR images due to the presence of speckle noise and irregular patterns, which significantly hindered segmentation performance. In contrast, the SynEO images led to successful segmentation by the

130 Segment Anything model. Although the results were still
131 not on par with the EO images, the performance showed
132 substantial qualitative improvement. Improving the SAR-
133 to-EO translation quality could enable generalized models
134 like Segment Anything to be more effectively utilized on
135 SAR data.

136 D. Limitations and Border Impact

137 D.1. Limitations and future works

138 While SAR images may appear visually similar across re-
139 gions, their actual representations can differ significantly
140 due to variations in surface roughness. Therefore, regional
141 characteristics play a critical role in SAR-to-EO translation.
142 However, in this work, we do not leverage extra modalities,
143 such as OSM-based label maps or metadata (longitude, lati-
144 tude, ground sample distance, and date), which can be used
145 to better account for these regional differences.

146 The SAR2Earth dataset does provide metadata informa-
147 tion for each image pixel level, allowing for future research
148 to utilize this additional data. As part of our future work,
149 we plan to incorporate extra modality-based regional fea-
150 tures to enhance the performance of SAR-to-EO translation
151 models by making them more sensitive to regional varia-
152 tions.

153 D.2. Border Impact

154 Our work holds significant implications for remote sensing
155 and related fields. By providing a public benchmark dataset,
156 we aim to accelerate research in SAR-to-EO translation, fa-
157 cilitating advancements in environmental monitoring, dis-
158 aster response, and urban development. SAR2Earth is par-
159 ticularly valuable in disaster scenarios like floods, where
160 heavy cloud cover renders traditional EO imagery less ef-
161 fective. Since SAR can penetrate clouds and is unaffected
162 by weather conditions or daylight, translating SAR-to-EO
163 images can provide critical information when it is most
164 needed. By translating SAR images into EO-like images,
165 we facilitate the application of advanced AI models devel-
166 oped for EO imagery to SAR data, potentially maximizing
167 the utility of established methodologies.

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